

HARDWARE POWER ELECTRONICS PROJECT TOPICS

1. MPPT Control of Solar PV Systems Using Buck Converter with Perturb & Observe Algorithm
2. Fuzzy Logic-Based MPPT with Cuk Converter for Solar Energy Systems
3. Neural Network MPPT Control for Photovoltaic Systems Using Buck-Boost Converter
4. Incremental Conductance MPPT Implementation in Solar PV with SEPIC Converter
5. Particle Swarm Optimization-Based MPPT for PV Systems Using LUO Converter
6. Comparative Study of DC-DC Converters in MPPT Control of Solar Photovoltaic Systems
7. Hybrid MPPT Algorithm for Solar PV Integrated with CUK and Buck Converters
8. Sliding Mode Control for MPPT in PV Systems Using Interleaved Boost Converter
9. Design and Implementation of MPPT with SEPIC-LUO Combined Converter in PV Applications
10. Design and Hardware Co-Simulation of a Bidirectional Buck-Boost Converter for Grid-Connected Energy Storage Systems
11. Design and Implementation of a Variable Voltage Buck-Boost Converter for Portable Electronic Devices
12. High-Efficiency Cuk Converter Design for Variable Power Supply in Portable Gadgets
13. Development of a Compact SEPIC Converter for Voltage Regulation in Handheld Devices
14. Design and Control of a Zeta Converter for Battery-Powered Portable Electronics
15. Implementation of a Bidirectional Buck Converter for Smart Charging in Portable Power Banks
16. Design of a Synchronous Boost Converter with Dynamic Voltage Scaling for Wearable Electronics
17. Modeling and Simulation of a Flyback Converter for Portable Medical Devices

18. Ultra-Low Power Buck Converter Design for IoT-Based Portable Sensor Nodes
19. Design of a Multi-Output Interleaved Boost Converter for Portable Communication Systems
20. High-Frequency Luo Converter for Fast-Charging Portable Consumer Electronics
21. Design and Implementation of a High-Efficiency SEPIC Converter for Solar Charging
22. Multi-Output Buck Converter for Low-Power Embedded Systems
23. Simulation of a Single-Input Multiple-Output Luo Converter for Distributed Loads
24. Cascaded Boost-Buck Converter Design for Portable Power Units
25. Implementation of a High Step-Up Interleaved Boost Converter for Solar-Powered Inverters
26. Analysis of a Modified Cuk Converter with Low Ripple for Renewable Systems
27. Flyback Converter-Based USB Charging System for IoT Devices
28. SEPIC-Luo Hybrid Converter for High Voltage Gain Applications
29. Adaptive PWM Control in Buck-Boost Converter for Dynamic Loads
30. Zeta Converter Design for Medical Monitoring Devices
31. Intelligent MPPT Controller Using Fuzzy Logic with Cuk Converter
32. High-Efficiency Solar Charge Controller with Modified SEPIC and Incremental Conductance MPPT
33. MPPT in Hybrid Wind-Solar Systems Using Interleaved Boost Converter
34. PV-Based Water Pumping System Using Buck Converter and P&O MPPT
35. Real-Time MPPT Tracking Using ANN with SEPIC Converter
36. Solar Energy Harvesting System Using Bidirectional Boost Converter
37. Solar-Powered Street Lighting System with Cuk Converter and MPPT
38. Design of a Dual-Input MPPT Controller for Solar-Wind Hybrid Systems
39. Digital MPPT Implementation for Portable PV Systems Using Zeta Converter

40. MPPT Using Genetic Algorithm in Solar Battery Chargers with Buck-Boost Converter
41. Design of a Bidirectional Flyback Converter for EV Regenerative Braking
42. Wireless Power Transfer System for EV Charging Using Resonant Converter
43. Modular EV Battery Charger Using Bidirectional Buck Converter
44. Intelligent Charging of Electric Scooters Using SEPIC Converter
45. Regenerative Braking in E-Rickshaws Using Bidirectional Cuk Converter
46. Adaptive Power Management in Electric Vehicles Using Multi-Mode DC-DC Converters
47. EV Battery Swapping Station with Flyback Converter and Energy Balancing
48. Fast EV Charging System Using Interleaved Boost Converter with Digital Control
49. Hybrid Supercapacitor-EV Battery System Using Buck-Boost Converter
50. Design of Solar EV Charging Station with SEPIC Converter and MPPT
51. Renewable Energy-Based Smart Refrigerator Using Cuk Converter
52. Intelligent Fan Speed Controller Using Buck Converter and Temperature Sensing
53. Smart Power Supply for IoT Gateways Using Zeta Converter
54. Home Automation Power Hub Using Single Input, Multi-Output SEPIC Converter
55. Battery-Powered Home UPS Using Bidirectional Boost Converter
56. Low-Cost MPPT Charge Controller for Rural Electrification
57. Multi-Device Charging Station Using Modified Buck Converter
58. Solar-Powered EV Surge Protection Using Flyback Converter
59. Energy-Efficient Lighting System Using SEPIC Converter and Occupancy Sensor
60. Portable Solar Generator with MPPT and Multi-Port Outputs
61. Adaptive PID Control of Flyback Converter for Power Management
62. Sliding Mode Control of a Bidirectional Buck-Boost Converter in EVs
63. ANN-Based Digital Control of High-Frequency Luo Converter

64. Control-Oriented Design of Interleaved Boost Converter for Renewable Systems
65. Simulation of PWM-Based Cuk Converter Using MATLAB Simulink
66. Predictive Control of Buck Converter for Dynamic Load Regulation
67. IoT-Enabled Monitoring and Control of Solar Charge Controllers (ANN, P&O)
68. Design and Implementation of an Interleaved SEPIC Converter for High-Efficiency PV Systems
69. Interleaved Boost Converter for High Power Density Applications in Electric Vehicles
70. Interleaved Luo Converter for Ripple Reduction in Renewable Energy Storage Systems
71. High-Performance Interleaved Boost Converter for Fuel Cell-Based Power Supply
72. Adaptive Control of Interleaved SEPIC Converter for Low Input Voltage Applications
73. Design and Simulation of an Isolated Flyback Converter for Battery-Powered Devices
74. Digital Control of a High-Frequency Isolated Flyback Converter for Medical Equipment
75. Flyback Converter with Galvanic Isolation for Solar Micro-Inverter Applications
76. Isolated Flyback Converter for Portable Power Banks with Fast Charging Capability
77. Design and Implementation of an Isolated Flyback Converter with Primary-Side Regulation
78. Bridgeless Cuk Converter for Power Factor Correction in AC-DC Converters
79. High-Efficiency Bridgeless SEPIC Converter for Solar Battery Charging Applications
80. Bridgeless Boost Converter for Low THD and High Power Factor in Front-End Rectifiers
81. Design of a Bridgeless SEPIC Converter with MPPT for PV Energy Systems
82. Bidirectional Bridgeless Cuk Converter for Electric Vehicle Onboard Chargers
83. Interleaved Bridgeless Boost Converter for EV Fast Charging Stations
84. MPPT-Controlled Bridgeless SEPIC Converter for Solar-Powered Smart Appliances

85. Bridgeless SEPIC-Luo Hybrid Converter for Off-Grid Power Systems
86. Low-Loss Bridgeless Cuk Converter for Wireless Power Transfer Systems
87. Bridgeless Boost Converter with Synchronous Rectification for Power Factor Correction
88. Design and Implementation of a Bidirectional DC-DC Converter for V2G Power Transfer in Electric Vehicles
89. Smart Control of Bidirectional Boost Converter for V2G Integration in Smart Grids
90. Grid-Tied V2G System Using Bidirectional Cuk Converter with Power Factor Correction
91. Battery-to-Grid Energy Discharge Using Isolated Bidirectional Flyback Converter in V2G Applications
92. Adaptive Fuzzy Logic-Based Control of Bidirectional Converter for Efficient V2G Operation
93. Development of a Bidirectional DC-DC Converter for V2V Energy Sharing Between Electric Vehicles
94. Design of a Compact Bidirectional SEPIC Converter for Vehicle-to-Vehicle Charging Systems
95. Dynamic Load Management in V2V Systems Using Bidirectional Buck-Boost Converter
96. Multi-Mode Bidirectional Power Converter for Combined V2G and V2V Applications in Smart Transportation Systems